Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CE711]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

ADVANCED CONCRETE TECHNOLOGY

Time: 3 Hours **Maximum Marks: 70**

Instructions:

i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Use of **IS 10262-2016** code is permitted.

iii. Missing data may be suitably assumed.

1. a. What are bouge's compound? Explain briefly different bogue's compounds and its effect on concrete. **07**

b. Discuss the factors affecting the Modulus of Elasticity. **07**

2. a. What are new generation super plasticizers? Explain the working mechanism with reference to sketches **07**

b. Explain briefly the role of GGBS and it's effect on concrete property in fresh and hardened state. **07**

3. Calculate the mix design for M 30 grade of concrete **14**

Type of cement –OPC 53

Maximum size of aggregate -20 mm

Exposure conditions- Moderate (RCC)

Workability -25-50mm slump

Minimum cement content -350 kg/m3

Maximum cement content: 450 kg/m3

Maximum W/C -0.5

Method of placing concrete- pumping

Degree of supervision-Good

Type of aggregate -crushed angular aggregate

Super plasticizer will be used

Specific gravity of coarse aggregate -2.68

Specific gravity of fine aggregate -2.65

Water absorption coarser aggregate-0.6%, fine aggregates-1%

Grading of coarse aggregates is conforming to table 2 of IS 383 and grading of fine grading of coarse aggregates is conforming to table 2 of IS 383 and grading of fine aggregate is falling in zone I.

OR

4.		Design a concrete mix for construction of an elevated water tank. The specified design strength of concrete (characteristic strength) is 30 MPa at 28 days measured on standard cylinders. Standard deviation can be taken as 4 MPa. The specific gravity of FA and C.A. are 2.65 and 2.7 respectively. The dry rodded bulk density of C.A. is 1600 kg/m3, and fineness modulus of FA is 2.80. Ordinary Portland cement (Type I) will be used. A slump of 50 mm is necessary, Assume any suitable missing data.as per ACI Method	**14**
5.	**a.**	What is Freezing & Thawing in concrete? List out the causes and preventive measures.	**07**
	b.	Briefly Explain the ultrasonic pulse velocity test on concrete with working principle.	**07**
		OR	
6.	**a.**	Explain the reasons how the durability of concrete is reduced due to sulphate attack. What are the counter measure to reduce it?.	**07**
	b.	Briefly explain the measurement of reinforcement and cover in structural element by using profometer.	**07**
7.	**a.**	What are the basic properties of fiber – reinforced concrete which can be advantageously made use of in the design of structural elements?.	**07**
	b.	What is underwater construction? Explain various methods of underwater construction.	**07**
		OR	
8.	**a.**	Explain the advantages of the prefabrication technology.	**07**
	b.	Describe the procedure of slip form construction.	**07**